Practical: 07

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

CODE:

```
~

user@AnushkaGudadhe MSYS ~
$ nano lru.c
```

```
~
  GNU nano 8.7
#include <stdio.h>

int main() {
    int frames[3];
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int time[3];
    int i, j, k;
    int hit = 0, fault = 0;
    int n = 12, frames_count = 3;
    int counter = 0, flag;

    // initialize frames
    for(i = 0; i < frames_count; i++) {
        frames[i] = -1;
    }

    for(i = 0; i < n; i++) {
        flag = 0;

        // check page hit
        for(j = 0; j < frames_count; j++) {
            if(frames[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                flag = 1;
                break;
            }
        }

        // page fault
        if(flag == 0) {
            int min = 0;

            for(j = 1; j < frames_count; j++) {
                if(time[j] < time[min])
                    min = j;
            }

            frames[min] = pages[i];
            counter++;
            time[min] = counter;
            fault++;
        }
    }

    float hit_percent = (hit * 100.0) / n;
    float fault_percent = (fault * 100.0) / n;
```

```
        // page fault
        if(flag == 0) {
            int min = 0;

            for(j = 1; j < frames_count; j++) {
                if(time[j] < time[min])
                    min = j;
            }

            frames[min] = pages[i];
            counter++;
            time[min] = counter;
            fault++;
        }
    }

    float hit_percent = (hit * 100.0) / n;
    float fault_percent = (fault * 100.0) / n;

    printf("Total Page Hits: %d\n", hit);
    printf("Total Page Faults: %d\n", fault);
    printf("Page Hit Percentage: %.2f%%\n", hit_percent);
    printf("Page Fault Percentage: %.2f%%\n", fault_percent);

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line
```

OUTPUT:

```
user@AnushkaGudadhe MSYS ~
$ gcc lru.c -o lru

user@AnushkaGudadhe MSYS ~
$ ./lru
Total Page Hits: 2
Total Page Faults: 10
Page Hit Percentage: 16.67%
Page Fault Percentage: 83.33%

user@AnushkaGudadhe MSYS ~
$ |
```